1
Past
The Emergence of AI

1.0 Summary

Section 1.1 begins by offering a brief overview of how digital developments have led to the current availability and success of AI systems. Section 1.2 interprets the disruptive impact of digital technologies, sciences, practices, products, and services—in short, *the digital*—as being due to its ability to cut and paste realities and ideas that we have inherited from modernity. I call this the *cleaving power* of the digital. I illustrate this cleaving power through some concrete examples. Then I use it to interpret AI as a new form of *smart agency* brought about by the digital decoupling of agency and intelligence, an unprecedented phenomenon that has caused some distractions and misunderstandings such as 'the Singularity', for example. Section 1.3 presents a short excursus into political agency, the other significant kind of agency that is transformed by the cleaving power of the digital. It briefly explains why this topic is essential and highly relevant, yet also lies beyond the scope of this book. Section 1.4 returns to the main issue of a conceptual interpretation of AI and introduces Chapter 2 by reminding the reader of the difficulty of defining and characterizing precisely what AI is. In the concluding section, I argue that *design* is the counterpart to the cleaving power of the digital and anticipate some of the topics discussed in the second half of the book.

1.1 Introduction: The Digital Revolution and AI

In 1964 Paramount Pictures distributed *Robinson Crusoe on Mars*. The movie described the adventures of Commander Christopher 'Kit' Draper (Paul Mantee), an American astronaut shipwrecked on Mars. Spend just a few minutes watching it on YouTube, and you will see how radically the world has changed in just a few decades. In particular, the computer at the very beginning of the movie looks like a Victorian engine with levers, gears, and dials—a piece of archaeology that Dr Frankenstein might have used. And yet, towards the end of the story, Friday (Victor Lundin) is tracked by an alien spacecraft through his bracelets, a piece of futurology that seems unnervingly prescient.

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Robinson Crusoe on Mars belonged to a different age, technologically and culturally closer to the preceding century than to ours. It describes a *modern*, not *contemporary*, reality based on *hardware*, not *software*. Laptops, the Internet, web services, touch screens, smart phones, smart watches, social media, online shopping, video and music streaming, driverless cars, robotic mowers, virtual assistants, and the Metaverse were all still to come. AI was mainly a project, not a reality. The movie shows technology made of nuts and bolts, and mechanisms following the clunky laws of Newtonian physics. It was an entirely analogue reality based on atoms rather than bytes. This reality is one that millennials have never experienced, having been born after the early 1980s. To them, a world without digital technologies is like what a world without cars was for me (coincidentally, I was born in 1964): something I had only heard described by my grandmother.

It is often remarked that a smart phone packs far more processing power in a few inches than NASA could put together when Neil Armstrong landed on the moon five years after *Robinson Crusoe on Mars*, in 1969. We have all this power at an almost negligible cost. For the fiftieth anniversary of the moon landing in 2019, many articles ran comparisons and here are some staggering facts. The Apollo Guidance Computer (AGC) on board of Apollo 11 had 32,768 bits of random-access memory and 589,824 bits (72 KB) of read-only memory. You could not have stored this book on it. Fifty years later, your average phone came with 4 GB of RAM and 512 GB of ROM. That is about 1 million times more RAM and 7 million times more ROM. As for the processor, the AGC ran at 0.043 MHz. An average iPhone processor is said to run at about 2490 MHz, which is about 58,000 times faster. To get a better sense of the acceleration, perhaps another comparison may help. On average, a person walks at a pace of 5 km/h. A hypersonic jet travels slightly more than a thousand times faster at 6,100 km/h, just over five times the speed of sound. Imagine multiplying that by 58,000.

Where did all this speed and computational power go? The answer is twofold: *feasibility* and *usability*. In terms of applications, we can do more and more. We can do so in increasingly easy ways—not only in terms of programming but above all in terms of user experience. Videos, for example, are computationally very hungry. So are operating systems. AI is possible today also because we have the computational power required to run its software.

Thanks to this mind-boggling growth in storage and processing capacities at increasingly affordable costs, billions of people are connected today. They spend many hours online daily. According to Statista.com, for example, 'as of 2018, the average time spent using the internet [in the UK] was 25.3 hours per week. That was an increase of 15.4 hours compared to 2005.'[1] This is far from unusual, and the pandemic has made an even more significant difference now. I shall return to

[1] https://www.statista.com/statistics/300201/hours-of-internet-use-per-week-per-person-in-the-uk/.

this point in Chapter 2, but another reason why AI is possible today is because we are increasingly spending time in contexts that are digital and AI-friendly.

More memory, power, speed, and digital environments and interactions have generated more data in immense quantities. We have all seen diagrams with exponential curves indicating amounts that we do not even know how to imagine. According to the market intelligence company IDC,[2] the year 2018 saw humanity reach 18 zettabytes of data (whether created, captured, or replicated). The astonishing growth in data shows no sign of slowing down; apparently, it will reach 175 zettabytes in 2025. This is hard to grasp in terms of quantity, but two consequences deserve a moment of reflection.[3] First, the speed and the memory of our digital technologies are not growing at the same pace as the universe of data. So, we are quickly moving from a culture of recording to one of deleting; the question is no longer what to save, but what to delete to make room for new data. Second, most of the available data have been created since the nineties (even if we include every word uttered, written, or printed in human history and every library or archive that ever existed). Just look at any of those online diagrams illustrating the data explosion: the amazing side is not only on the right, where the arrow of growth goes, but also on the left, where it starts. It was only a handful of years ago. Because all the data we have were created by the current generation, they are also aging together in terms of support and obsolete technologies. So, their curation will be an increasingly pressing issue.

More computational power and more data have made the shift from logic to statistics possible. Neural networks that were once only theoretically interesting[4] have become common tools in machine learning (ML). Old AI was primarily symbolic and could be interpreted as a branch of mathematical logic, but the new AI is mostly connectionist and can be interpreted as a branch of statistics. The main battle horse of AI is no longer logical deduction, but statistical inference and correlation.

Computational power and speed, memory size, data volumes, the effects of algorithms and statistical tools, and the number of online interactions have all been growing incredibly fast. This is also because (here, the causal connection goes both ways) the number of digital devices interacting with each other is already several times higher than the human population. So, most communication is now machine to machine with no human involvement. We have computerized robots on Mars remotely controlled from Earth. Commander Christopher 'Kit' Draper would have found them utterly amazing.

All the previous trends will keep growing, relentlessly, for the foreseeable future. They have changed how we learn, play, work, love, hate, choose, decide,

[2] See the discussion in https://www.seagate.com/gb/en/our-story/data-age-2025/.
[3] I discuss both in Floridi (2014a).
[4] I discussed some of them in a much earlier book from the late nineties (Floridi 1999).

produce, sell, buy, consume, advertise, have fun, care and take care, socialize, communicate, and so forth. It seems impossible to locate a corner of our lives that has not been affected by the digital revolution. In the last half century or so, our reality has become increasingly digital. It is made up of zeros and ones, and run by software and data rather than hardware and atoms. More and more people live increasingly *onlife* (Floridi 2014b), both online and offline, and in the infosphere, both digitally and analogically.

This digital revolution also affects how we conceptualize and understand our realities, which are increasingly interpreted in computational and digital terms. Just think of the 'old' analogy between your DNA and your 'code', which we now take for granted. The revolution has fuelled the development of AI as we share our onlife experiences and our infosphere environments with artificial agents (AA), whether algorithms, bots, or robots. To understand what AI may represent—I shall argue that it is a new form of agency, not of intelligence—more needs to be said about the impact of the digital revolution itself. That is the task of the rest of this chapter. It is only by understanding the conceptual trajectory of its implications that we can gain the right perspective about the nature of AI (Chapter 2), its likely developments (Chapter 3), and its ethical challenges (Part Two).

1.2 Digital's Cleaving Power: Cutting and Pasting Modernity

Digital technologies, sciences, practices, products, and services—in short, *the digital* as an overall phenomenon—are deeply transforming reality. This much is obvious and uncontroversial. The real questions are *why*, *how*, and *so what*, especially as they relate to AI. In each case, the answer is far from trivial and certainly open to debate. To explain the answers I find most convincing, and thus introduce an interpretation of AI as an increasing *reservoir of smart agency* in the next chapter, let me start *in medias res*, that is, from the 'how'. It will then be easier to move backwards to understand the 'why' and then forward to deal with the 'so what' before linking the answers to the emergence of AI.

The digital 'cuts and pastes' our realities both ontologically and epistemologically. By this I mean that it couples, decouples, or recouples features of the world (ontology) and therefore our corresponding assumptions about them (epistemology) that we thought were unchangeable. It splits apart and fuses the 'atoms' of our 'modern' experience and culture, so to speak. It reshapes the bed of the river, to use a Wittgensteinian metaphor. A few stark examples may bring home the point vividly.

Consider first one of the most significant cases of coupling. Self-identity and personal data have not always been glued together as indistinguishably as they are today, when we speak of the *personal identity* of *data subjects*. Census counts are very old (Alterman 1969). The invention of photography had a huge impact on privacy (Warren and Brandeis 1890). European governments made travelling with a passport compulsory during the First World War, for migration and security reasons, thus extending the state's control over the means of mobility (Torpey 2000). But it is only the digital, with its immense power to record, monitor, share, and process boundless quantities of data about Alice, that has soldered together who Alice is, her individual self and profile, with personal information about her. Privacy has become a pressing issue also, if not mainly, because of this coupling. Today, in EU legislation at least, data protection is discussed in terms of human dignity (Floridi 2016c) and personal identity (Floridi 2005a, 2006, 2015b) with citizens described as *data subjects*.

The next example concerns *location* and *presence* and their decoupling. In a digital world, it is obvious that one may be physically located in one place (say, a coffee shop) and interactively present in another (say, a page on Facebook). Yet all past generations that lived in an exclusively analogue world conceived and experienced location and presence as two inseparable sides of the same human predicament: being situated in space and time, here and now. Action at a distance and telepresence belonged to magical worlds or science fiction. Today, this decoupling simply reflects ordinary experience in any information society (Floridi 2005b). We are the first generation for which 'where are you?' is not just a rhetorical question. Of course, the decoupling has not severed all links. Geolocation works only if Alice's telepresence can be monitored. And Alice's telepresence is possible only if she is located within a physically connected environment. But the two are now completely distinguished and, indeed, their decoupling has slightly downgraded location in favour of presence. Because if all Alice needs and cares about is to be digitally present and interactive in a particular corner of the infosphere, it does not matter where in the world she is located analogically, whether at home, on a train, or in her office. This is why banks, bookstores, libraries, and retail shops are all *presence* places in search of a *location* repurposing. When a shop opens a cafeteria, it is trying to paste together the presence and the location of customers—a connection that has been cut by the digital experience.

Consider next the decoupling of *law* and *territoriality*. For centuries, roughly since the 1648 Peace of Westphalia, political geography has provided jurisprudence with an easy answer to the question of how widely a ruling should apply: it should apply as far as the national borders within which the legal authority operates. That coupling could be summarized as 'my place, my rules; your place, your rules'. It may now seem obvious, but it took a long time and immense suffering to reach such a simple approach. The approach is still perfectly fine today, if one is operating only within a physical, analogue space. However, the Internet is not a physical space. The territoriality problem arises from an ontological misalignment between the normative space of law, the physical space of geography, and the logical space of the digital. This is a new, variable 'geometry' that we are still learning to manage. For instance, the decoupling of law and territoriality became obvious as well as problematic during the debate on the so-called *right to be forgotten* (Floridi 2015a). Search engines operate within an online logical space of nodes, links, protocols, resources, services, URLs (uniform resource locators), interfaces, and so forth. This means anything is only a click away. So, it is difficult to implement the right to be forgotten by asking Google to remove links to someone's personal information from its .com version in the United States due to a decision taken by the Court of Justice of the European Union (CJEU), even if that decision may seem pointless, unless the links are removed from all versions of the search engine.

Note that such misalignment between spaces both causes problems and provides solutions. The non-territoriality of the digital works wonders for the unobstructed circulation of information. In China, for example, the government must make a constant and sustained effort to control information online. Along the same lines, the General Data Protection Regulation (GDPR) must be admired for its ability to exploit the 'coupling' of personal identity and personal information to bypass the 'decoupling' of law and territoriality. It does so by grounding the protection of personal data in terms of the former (to whom they are 'attached', which is now crucial) rather than the latter (where they are being processed, which is no longer relevant).

Finally, here is a coupling that turns out to be more accurately a recoupling. In his 1980 book *The Third Wave*, Alvin Toffler coined the term 'prosumer' to refer to the blurring and merging of the role of producers and consumers (Toffler 1980). Toffler attributed this trend to a highly saturated marketplace and mass of standardized products prompting a process of mass customization, which in turn led to the increasing involvement of consumers as producers of their own customized products. The idea was anticipated in 1972 by Marshall McLuhan and Barrington Nevitt, who attributed the phenomenon to electricity-based technologies. It later came to refer to the consumption of information produced by the same population of producers, such as on YouTube. Unaware of these precedents, I introduced the word 'produmer' to capture the same phenomenon almost twenty years after Toffler.[5] But in all these cases, what is at stake is not a *new* coupling. More precisely, it is a recoupling.

For most of our history (approximately 90 per cent of it; see Lee and Daly (1999)), we have lived in hunting-gathering societies, foraging to survive. During this length of time, producers and consumers normally overlapped. Prosumers hunting wild animals and gathering wild plants were, in other words, the normality rather than the exception. It is only since the development of agrarian societies, around ten thousand years ago, that we have seen a complete (and after a while

[5] In Floridi (1999); see also Floridi (2004, 2014b). I should have known better and used Toffler's 'prosumer'.

culturally obvious) separation between producers and consumers. But in some corners of the infosphere, that decoupling is being recoupled. On Instagram, TikTok, or Clubhouse, for example, we consume what we produce. One may therefore stress that, in some cases, this parenthesis is coming to an end and that prosumers are back, recoupled by the digital. Consequently, it is perfectly coherent that human behaviour online has been compared and studied in terms of foraging models since the 1990s (Pirolli and Card 1995, 1999, Pirolli 2007).

The reader may easily list more cases of coupling, decoupling, and recoupling. Think, for example, about the difference between *virtual reality* (decoupling) and *augmented reality* (coupling); the ordinary decoupling of *usage* from *ownership* in the share economy; the recoupling of *authenticity* and *memory* thanks to blockchain; or the current debate about a universal basic income, which is a case of decoupling *salary* from *work*. But it is time to move from the 'how' to the 'why' question. Why does the digital have this *cleaving power*[6] to couple, decouple, and recouple the world and our understanding of it? Why do other technological innovations seem to lack a similar impact? The answer, I surmise, lies in the combination of two factors.

On the one hand, the digital is a *third-order technology* (Floridi 2014a). It is not just a technology between us and nature, like an axe (first order), or a technology between us and another technology, like an engine (second order). It is rather a technology between one technology and another technology, like a computerized system controlling a robot painting a car (third order). Due to the autonomous processing power of the digital, we may not even be on—let alone in—the loop.

On the other hand, the digital is not merely enhancing or augmenting reality. It radically transforms reality because it creates new environments that we then inhabit, and new forms of agency with which we then interact. There is no term for this profound form of transformation. In the past (Floridi 2010b), I used the expression *re-ontologizing* to refer to a very radical kind of re-engineering. This sort of re-engineering not only designs, constructs, or structures a system (e.g. a company, a machine, or some artefact) anew but fundamentally transforms the intrinsic nature of the system itself—that is, its ontology. In this sense, nanotechnologies and biotechnologies are not merely re-engineering but re-ontologizing our world. And by *re-ontologizing modernity*, to put it shortly, the digital is *re-epistemologizing modern mentality* (that is, many of our old conceptions and ideas) as well.

[6] I have chosen 'cleaving' as a particularly apt term to refer to the de/recoupling power of the digital because 'to cleave' has two meanings: (a) 'to split or sever something', especially along a natural line or grain; and (b) 'to stick fast or adhere strongly to' something. This may seem contradictory, but it is due to the fact that 'to cleave' is the result of a merging into a single spelling, and hence into a twofold meaning, of two separate words: (a) comes from the Old English cleofan, related to the German klieben (to cut through); whereas (b) comes from the Old English clifian, related to the German kleben (to stick or cling). They have very different Proto-Indo-European roots; see http://www.etymonline.com/index.php?term=cleave.

Put together, all these factors suggest that the digital owes its cleaving power to being a re-ontologizing, re-epistemologizing, third-order technology. This is why it does what it does, and why no other technology has come close to having a similar effect.

1.3 New Forms of Agency

If all this is approximately correct, then it may help one make sense of some current phenomena concerning the transformation of the morphology of agency in the digital age, and hence about *forms of agency* that only seem unrelated. Their transformation depends on the cleaving power of the digital, but their interpretation may be due to an implicit misunderstanding about this cleaving power and its increasing, profound, and long-lasting consequences. I am referring to *political agency* as direct democracy and *artificial agency* as AI. In each case, the re-ontologizing of agency has not yet been followed by an adequate re-epistemologizing of its interpretation. Or to put it less precisely but perhaps more intuitively: the digital has changed the nature of agency, but we are still interpreting the outcome of such changes in terms of modern mentality, and this is generating some deep misunderstanding.

The kind of agency I am referring to is not the one most discussed in philosophy or psychology, involving mental states, intentionality, and other features that are typically associated to human beings. The agency discussed in this book is the one that is commonly found in computer science (Russell and Norvig 2018) and in the literature on multi-agent systems (Weiss 2013, Wooldridge 2009). It is more minimalist and requires that a system satisfies only three basic conditions: it can

a) receive and use data from the environment, through sensors or other forms of data input;
b) take actions based on the input data, autonomously, to achieve goals, through actuators or other forms of output, and
c) improve its performance by learning from its interactions.

A similar agent can be artificial (e.g. a bot), biological (e.g. a dog), social (e.g. a company, or a government), or hybrid. In what follows, I shall say something very brief about political agency and direct democracy because this book will concentrate only on artificial agency, not socio-political agency shaped and supported by digital technologies.

In current debates about direct democracy, we are sometimes misled into thinking that the digital *should* (note the normative as opposed to descriptive approach) recouple *sovereignty* and *governance*. Sovereignty is political power that can be legitimately delegated, while governance is political power that is legitimately delegated—on a temporary, conditional, and accountable basis—and can be equally legitimately withdrawn (Floridi 2016e). *Representative* democracy is commonly, if mistakenly, seen as a compromise due to practical communication constraints. True democracy would be *direct*, based on the unmediated, constant, and universal participation of all citizens in political matters. Unfortunately, so goes the argument, we are too many. So the delegation (understood as intermediation) of political power is a necessary, if minor, evil (Mill (1861), 69). It is the myth of the city-state and especially of Athens.

For centuries, the compromise in favour of a representative democracy has seemed inevitable—that is, until the arrival of the digital. According to some people, this now promises to disintermediate modern democracy and couple (or recouple, if you believe in some mythical good old days) sovereignty with governance to deliver a new sort of democracy. This would be some sort of digital agora that could finally enable the regular, direct involvement of every interested citizen. It is the same promise made by the referendum instrument, especially when it is binding rather than advisory. In either case, voters are asked directly what should be done. The only task left to the political, administrative, and technical class would be that of implementing people's decisions. Politicians would be *delegated* (not *representative*) civil servants in a very literal sense. Yet this is a mistake because indirect democracy was always the plan. To be trite for a moment, the decoupling is a feature rather than a bug. A democratic regime is characterized above all not by some *procedures* or some *values* (although these can also be characteristics) but by a clear and neat *separation*—that is, a decoupling—between those who hold political power (sovereignty) and those who are entrusted with it (governance). All voting-age citizens have political power and delegate it legitimately by voting. This power is then entrusted to others who exercise that mandate by governing transparently and accountably for as long as they are legitimately empowered. To put it simply, a democratic regime is not merely a way of exercising and managing power in some ways (procedures) and/or according to some values. It is, first and foremost, a way of *structuring* power. Those who hold political power do not exercise it; they give it to those who exercise it, but do not hold it. The conflation of the two sides leads to brittle forms of autocracy or mob rule.

From such a perspective, representative democracy is not a compromise. It is actually the best form of democracy. And using the digital to couple (or, as I have already remarked, more mythically recouple) sovereignty and governance would be a costly mistake. Brexit, Trump, Lega Nord and other populist disasters caused by the 'tyranny of the majority' (Adams 1787) are sufficient evidence. We need to consider how best to take advantage of the planned, representative decoupling between sovereignty and governance—not how to erase it. So, consensus is the problem. However, it is not a topic of this book. All I want to offer with the previous analysis is a taste of the sort of unifying considerations about forms of agency that link the impact of the digital on politics to how we think about and evaluate AI, as we shall see in the next section. Let us now return to artificial agency.

1.4 AI: A Research Area in Search of a Definition

Some people (perhaps many) seem to believe that AI is about coupling artificial agency and intelligent behaviour into new artefacts. This is a misunderstanding. As I shall explain at more length in the next chapter, the opposite is actually true: the digital revolution has made AI not only possible but also increasingly useful. It has done so by *decoupling* the ability to solve a problem or complete a task successfully from any need to be intelligent to do so. It is only when this decoupling is achieved that AI is successful. So, the usual complaint known as the 'AI effect'[7]—when as soon as AI can perform a particular task, such as automatic translation or voice recognition, the target then moves, and that task is no longer defined as intelligent if performed by AI—is actually a correct acknowledgement of the precise process in question. AI performs a task successfully only if it can decouple its completion from any need to be intelligent in doing so; therefore, if AI is successful, then the decoupling has taken place, and indeed the task has been shown to be decouplable from the intelligence that seemed to be required (e.g. in a human being) to lead to success.

This is less surprising than it may seem, and in the next chapter we shall see that it is perfectly consistent with the classic (and probably still one of the best) definitions of AI provided by McCarthy, Minsky, Rochester, and Shannon in their 'Proposal for the Dartmouth Summer Research Project on Artificial Intelligence', the founding document and later event that established the new field of AI in 1955. I shall only quote it here and postpone its discussion until the next chapter:

> for the present purpose the artificial intelligence problem is taken to be that of making a machine behave in ways that would be called intelligent if a human were so behaving.[8]

The consequences of understanding AI as a divorce between agency and intelligence are profound. So are the ethical challenges to which such a divorce gives rise, and the second part of the book is devoted to their analysis. But to conclude this introductory chapter, a final answer still needs to be provided for the 'so what' question I mentioned at the outset. This is the task of the next section.

1.5 Conclusion: Ethics, Governance, and Design

Assuming that the previous answers to the 'why' and 'how' questions are acceptable, what difference does it make when we understand the power of the digital in

[7] https://en.wikipedia.org/wiki/AI_effect.

[8] Online version at http://www-formal.stanford.edu/jmc/history/dartmouth/dartmouth.html; see also the reissue in McCarthy et al. (2006).

terms of cutting and pasting the world (along with our conceptualization of it) in unprecedented ways? An analogy may help introduce the answer. If one possesses only a single stone and absolutely nothing else, not even another stone to put next to it, then there is also nothing else one can do aside from enjoying the stone itself—perhaps by looking at it or playing with it. But if one cuts the stone into two, several possibilities emerge for combining them. Two stones provide more affordances and fewer constraints than a single stone, and many stones even more so. Design becomes possible.

Cutting and pasting the ontological and conceptual blocks of modernity, so to speak, is precisely what the digital does best when it comes to its impact on our culture and philosophy. And taking advantage of such affordances and constraints in view of solving some problems is called *design*. So, the answer should now be clear: the cleaving power of the digital hugely decreases constraints on reality and escalates its affordances. By doing this, it makes *design*—loosely understood as the art of solving a problem through the creation of an artefact that takes advantage of constraints and affordances, to satisfy some requirements, in view of a goal—the innovating activity that defines our age.

Our journey is now complete. Each age has innovated its cultures, societies, and environments by relying on at least three main elements: *discovery*, *invention*, and *design*. These three kinds of innovation are tightly intertwined, yet innovation has often been skewed like a three-legged stool in which one leg is longer and hence more forward than the others. The post-renaissance and early modern period may be qualified as the age of discovery, especially geographic. Late modernity is still an age of discovery but, with its industrial and mechanical innovations, is perhaps, even more, an age of invention. And, of course, all ages have also been ages of design (not least because discoveries and inventions require ingenious ways of linking and giving form to new and old realities). But if I am correct in what I have argued so far, then it is our age that is quintessentially, more so than any other, *the age of design*.

Because the digital is lowering the constraints and increasing the affordances at our disposal, it is offering us immense and growing freedom to arrange and organize the world in many ways to solve a variety of old and new problems. Of course, any design requires a project. And in our case, it is a *human project* for our digital age that we are still missing. But we should not let the cleaving power of the digital shape the world without a plan. We should make every effort to decide the direction in which we wish to exploit it, to ensure that the information societies we are building thanks to it are open, tolerant, equitable, just, and supportive of the environment as well as human dignity and flourishing. The most important consequence of the cleaving power of the digital should be a better design of our world. And this concerns the shaping of AI as a new form of agency, as we shall see in Chapter 2.